$$z=\sqrt{(5x+\tan(4x))}$$

$$z=(5x+\tan(4x))^{\frac{1}{2}}$$

$$\hat{z}=\frac{1}{2}(5x+\tan(4x))^{-\frac{1}{2}}.(5+4sec^2(4x))$$

$$\hat{z}=\frac{5+4sec^2(4x)}{2\sqrt{(5x+\tan(4x))}}$$